Table II: SysJava vs SWAT4J on the primary JDK 8 container images (Debian 10, Temurin 8u504).

	Zookeeper 3.4.14 JDK 8u504 Debian 10	Tomcat 7.0 JDK 8u504 Debian 10	Cassandra 3.0.29 JDK 8u504 Debian 10	Jetty 9.4.51 JDK 8u504 Debian 10	Storm 2.4.0 JDK 8u504 Debian 10	OrientDB 3.2.19 JDK 8u504 Debian 10	Elasticsearch 7.5.2 JDK 8u504 Debian 10	Solr 8.7.0 JDK 8u504 Debian 10	Groovy 4.0.0 JDK 8u504 Debian 10	JRuby 9.4.2.0 JDK 8u504 Debian 10
Native Method Resolution										
JNI Total Natives	495	816	975	353	1884	700	626	980	588	598
SysJava Resolved	484	792	903	352	1835	684	609	967	541	587
JNI Static (Java_*)	317	564	596	185	1347	498	427	645	360	416
Partial (_1/$/__)	4	59	31	2	65	18	14	8	7	6
JNI Dynamic (RegisterNatives)	119	122	226	119	366	123	121	261	122	120
JVM	44	47	45	46	47	45	47	47	52	45
JFR/Netty	0	0	5	0	10	0	0	6	0	0
Platform / Intrinsic	10	21	71	0	46	8	16	10	39	3
Not Found	1	3	1	1	3	8	1	3	8	8
SWAT4J Natives Found (JTracer)	236	614	686	225	1529	519	368	613	334	448
SWAT4J Resolved	129	433	408	124	283	351	174	298	174	330
SWAT4J JNI Static (Java_*)	93	394	264	86	140	313	135	154	133	291
JNI Partial/Heuristic	0	0	107	0	104	0	0	104	0	0
JVM Source Code	36	39	37	38	39	38	39	40	41	39
Not Found	107	181	278	101	1246	168	194	315	160	118
Syscall Allowlist (SysJava only)										
Java Lib Syscalls	113	128	127	100	133	115	118	131	113	103
Container Binary Syscalls	111	92	140	63	101	83	114	111	105	84
Go Whitelist (Δ)	36	35	24	50	27	40	32	28	37	42
Unmapped Lib Syscalls (Δ)	18	0	41	10	2	10	11	20	18	10
Total Syscalls	192	173	230	161	179	174	185	201	189	168
SWAT4J Syscall Comparison										
SWAT4J Java Lib Syscalls	108	123	119	108	118	109	109	118	108	105
SWAT4J Dynamic Syscalls	128	127	141	128	137	129	137	136	130	128
SWAT4J Total (Static ∪ Dynamic)	170	176	179	170	177	168	173	175	171	168
SysJava ∩ SWAT4J	141	145	162	134	146	136	143	150	141	136
SysJava only	51	28	68	27	33	38	42	51	48	32
SWAT4J only	29	31	17	36	31	32	30	25	30	32
Dynamic Capture Artefacts										
Libraries Loaded	20	24	119	15	29	23	35	28	19	19
Binaries Loaded	11	6	23	2	10	6	9	14	12	7
JARs Loaded	10	41	51	16	88	76	182	144	69	9
Directly Mapped Libraries Analysed	11	15	19	8	15	18	15	18	15	7

Note: SysJava Resolved + Platform / Intrinsic + Not Found = JNI Total Natives; SWAT4J Resolved + Not Found = SWAT4J Natives Found: SWAT4J maps the native methods its own bytecode analysis (JTracer) finds, by name, with the third-party JNI libraries shipped in its repository (it never reads the Hadoop and RocksDB ones). Java Lib ∪ Container Binary + Go Whitelist (Δ, runc entries not already counted) + Unmapped Lib (Δ, syscalls of loaded libraries without a Java binding, not already counted) = Total Syscalls. Native methods are found on an RTA call graph (virtual and interface calls dispatched to the overrides of every class instantiated in reachable code). Natives in libraries the image does not ship are mapped to the released library (Solr: Hadoop 3.2.0 libhadoop, Hadoop 3.3.6 libhadoop for the 14 erasure-coding natives, netty epoll 4.1.50, Lucene NativePosixUtil; OrientDB: jffi 1.3.11, NativePosixUtil; Elasticsearch: NativePosixUtil; Groovy: the jansi library bundled in jline 2.14.6; Tomcat: jsvc built from commons-daemon 1.2.4); each library is analysed once from all its mapped start functions. The remaining Not Found natives have no implementation in any library (e.g. jffi Foreign

natives never implemented upstream, JFR setCompressedIntegers absent from the 8u504 libjvm). Container binaries are analysed from _start and main; runc is covered by the Go whitelist only. Storm: sync in its generated profile comes only from FFM hits the detector inferred from Clojure class names (JDK 8 has no FFM API) and is not counted; Storm runs under the 179-syscall profile. Elasticsearch: SysPart cannot analyse the stripped ML controller; its syscalls are recovered from the functions it imports. SWAT4J Java Lib = its static allowlist (JTracer); SWAT4J Dynamic = strace of the container start-up. All ten containers run under their generated profiles.